ENGINEERING DESIGN REFERENCE · 12 PAGES

Every product.
Every number you need to start a design.

The full Precifarm range — home charging, fleet and highway charging, battery swap, modular energy and project storage — with ratings, sizing rules, installation requirements and approval gates in one place.

SPARK · PULSE · POD · DEPOT · CORRIDOR · BODA HUB · MODULAR ENERGY · MEGAPACK



The Precifarm range — portable Spark through highway Corridor DC, plus home storage and battery swap.

DOCUMENT	STATUS	COMPANIONS
PF-PRODUCT-ENG-001	Engineering Design Reference	PF-ENG-SOLAR-HUB-001 · PF-MODENERGY-002

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What Precifarm sells, and what is still concept

Status is part of the specification. On-sale products can be quoted and installed today. Conceptual products show design intent and must never be offered for sale.

Hardware portfolio

PRODUCT	ROLE	RATING	INTERFACE	STATUS	FROM
Spark charger	Portable AC	3.3 kW	Type 2	On sale	KES 25,000
Pulse charger	Home wallbox	7 kW AC	Type 2	On sale	KES 79,000
Pod energy storage	Home charger + battery	7 kW + 5–10 kWh	Type 2	On sale	KES 295,000
Depot charging station	Fleet / campus AC	22 kW AC	Type 2	On sale	Quoted per site
Corridor charging	Highway DC · Nairobi–Kisumu first	120 kW+ DC	Dual CCS2	On sale	KES 39/kWh
Boda Hub	Battery swap	2.56 kWh packs	Pack swap	Partner sites live	Fleet pricing
Automated Boda Hub	Self-service lockers	8–24+ doors	Pack swap	In design & testing	Not quoted
Energy Module	2.56 kWh building block	2.56 kW	Blind-mate	Conceptual	Not on sale
P1 Go	Portable backup	~1 kWh	UK 3-pin / USB	Conceptual	Not on sale
P2 Home	Home backup tower	2.6–10.2 kWh	Sub-board	Conceptual	Not on sale
Mini Stack	Outdoor SME backup	5.1–15.4 kWh	Sub-board	Conceptual	Not on sale
MegaPack	Project BESS	5 MWh – GWh	Project HV	Design-stage	Project-engineered

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Services and platform

SERVICE	WHAT IT DOES	STATUS
Charging Hub	Map of Precifarm DC, boda swap and partner stops — open vs coming soon	Live at /hub
Precifarm Agent	Web chat (DeepSeek V4 Flash) + OpenCode config + Android APK	Web live · OpenCode setup · Android early access
Engineering package	Design basis for home, depot, corridor and partner sites	Published (PF-ENG-SOLAR-HUB-001)
Training T1 / T2 / T3	Site staff, technician and commissioning on Pulse-Corridor hardware	Nairobi + corridor field modules
Partners & fleet	Hub hosting, fleet electrification, dealer and installer network	By contract
Lipa Pole Pole	M-Pesa instalment financing for Pulse and Pod only	Live · from KES 3,300/month
Charging-as-a-Service	Precifarm finances, installs and operates; pay per kWh or monthly	By contract

Evidence labels used across Precifarm documents

LABEL	MEANING
actual	Measured or delivered — verifiable today.
contracted	Committed in a signed agreement.
pipeline	In progress, not yet committed.
target	Design intent, not yet proven.
planning assumption	Used for sizing; confirm on site.
illustrative	Example figure to explain a concept.

Naming rules that must hold

Lipa Pole Pole is M-Pesa financing for Pulse and Pod — it is not a charger. Pod energy storage (home) is not Mini Stack (outdoor modular energy). MegaPack is project-engineered, not a catalogue item. Automated Boda Hub lockers are still in design and testing.

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Home charging — Spark, Pulse, Pod

Most Kenyan driving is a ~60 km day, about 10 kWh. That single number decides which product fits, and how long the vehicle needs to be plugged in.



Spark charger

Portable 3.3 kW

Boot-carried Type 2 lead for 13 A sockets. Top-up tool, not a daily home solution.



Pulse charger

Home 7 kW AC

Wall-mounted Type 2 wallbox. Consumer unit to parking bay, typically installed in one day.



Pod energy storage

7 kW + 5–10 kWh

7 kW charger plus 5 or 10 kWh LiFePO4 for weak-grid evenings. Essential-load backup — not whole-house air conditioning.

Home product specifications

PARAMETER	SPARK	PULSE	POD
Rated power	3.3 kW	7 kW	7 kW
Connector	Type 2	Type 2	Type 2
Supply	13 A socket	230 V / 50 Hz dedicated	230 V / 50 Hz dedicated
Storage	None	None	5 or 10 kWh LiFePO4
Typical day (~10 kWh / ~60 km)	~180 min	~90 min	~90 min
From (KES)	25,000	79,000	295,000
Lipa Pole Pole	No	Yes	Yes
Aftersale care	Warranty	3 years	3 years

Home energy packages — indicative, confirmed after survey

PACKAGE	FROM (KES)	CONTENTS	NOTES
Charge	120,000	7–11 kW AC wallbox, install	No solar or battery
Solar + Charge	450,000	4–5 kWp solar + 7–11 kW charger	Battery optional
Home Energy	650,000	~5 kWp + 5–10 kWh battery + 7–11 kW charger	Essential backup circuits

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Choosing between them

Spark covers a top-up where no wallbox exists. Pulse is the default for a home with a stable supply. Pod is for weak-grid sites where the customer also wants essential-load backup. Solar is added to Pod, not to Spark.

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What has to be true before install day

A home charger is a new continuous load on an existing installation. The survey exists to prove the supply, the board and the earthing can carry it.

Home survey and installation checklist

- **Supply check**
  Confirm main fuse, consumer-unit spare ways and incoming cable rating before quoting.
- **Dedicated circuit**
  Pulse and Pod run on their own protected radial — no spurs off socket circuits.
- **Protection**
  RCD Type A plus DC fault detection, or RCD Type B where the unit does not provide detection.
- **Earthing**
  Verify earthing arrangement and electrode resistance; PME export requires assessment.
- **Cable route**
  Size for load and length; check voltage drop at the far end of the run, not the origin.
- **Backup scope**
  Pod backs essential circuits on a sub-board — not air conditioning, cooking or water heating.
- **Commissioning**
  Load test, M-Pesa test session, remote monitoring registration, customer handover.



Home hybrid — grid, rooftop solar, storage and wallbox on one essential-load board.



Pod in a garage install — charger, storage and essential circuits in the same utility corner.

Running cost reference

ITEM	FIGURE	EVIDENCE
Home charging day (~60 km)	From KES 140	Illustrative · site tariff
Diesel equivalent day	From KES 1,000	Illustrative
Public DC session	From KES 39/kWh	Marketed floor · M-Pesa at bay
Boda swap	Under 5 minutes	Marketed at live swap sites

Financing

Lipa Pole Pole covers Pulse and Pod from KES 3,300 per month on M-Pesa. Deposit, monthly amount and total are shown before the customer commits. It does not apply to public charging sessions or Spark.

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Depot AC and Corridor DC

Depot suits vehicles that park for hours. Corridor suits vehicles that stop for minutes. Dwell time, not vehicle type, decides which one a site needs.



Depot — 22 kW AC (Pulse Plus pedestal) for fleet yards, campuses and partner retail.



Corridor — 120 kW+ DC, dual CCS2, M-Pesa tap-to-pay. First corridor: Nairobi–Kisumu.

Depot and Corridor specifications

PARAMETER	DEPOT STATION	CORRIDOR CHARGING
Rated power	22 kW AC	120 kW+ DC
Connector	Type 2	Dual CCS2
Typical session	40+ kWh in ~120 min	~60 kWh in 30 min
Design class	11–22 kW where dwell > 2 h	120–180 kW town · 240–320 kW highway
Form factor	Pedestal	T-canopy, overhead cables
Payment	M-Pesa · account billing	M-Pesa from KES 39/kWh
Typical host	Fleet yard, campus, partner retail	Fuel retail, highway stop, transport terminus

Five gates before Precifarm builds a hub

GATE	TEST
Proven demand	Enough EV sessions and corridor traffic to justify capital before deployment.
Power and tariff	Feeder study, interconnection and tariff clarity resolved before irreversible spend.
Site and dwell	Safe parking, shade and amenities — somewhere worth stopping, not just a plug.
Durable control	Long-term site rights at fuel retailers, malls, yards or transport termini.
Approvals	Regulatory and safety sign-off in place before public operation begins.

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Sizing a corridor site

Grid import is usually the binding constraint, not the charger. Storage closes the gap between what the vehicle wants in 30 minutes and what the feeder can supply.

Worked example — 120 kWh in 30 minutes

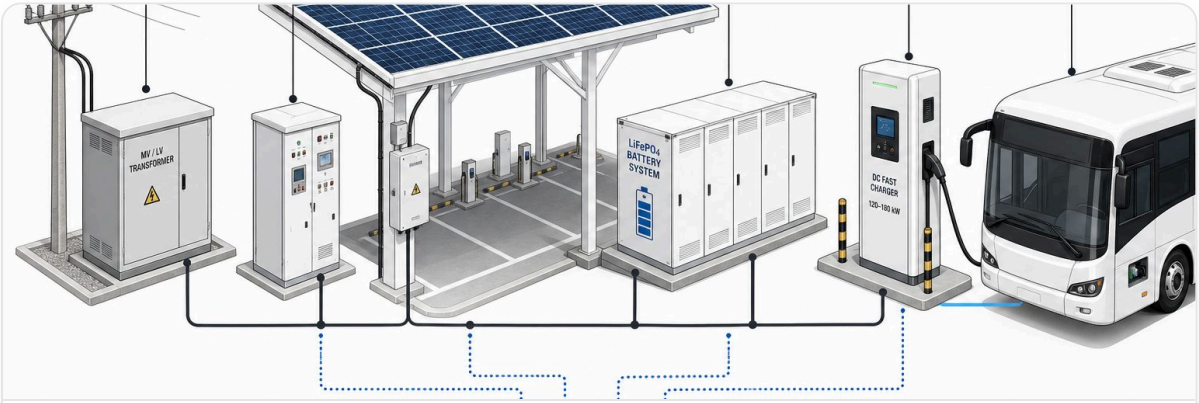
STEP	VALUE
Session target	120 kWh in 30 minutes
Power at vehicle	240 kW
Site demand with losses	~253 kW
Grid import limit	180 kW
Shortfall energy	~37 kWh per session
BESS sizing with reserve	80 – 100 kWh usable
Canopy PV	40 – 80 kWp → 180 – 440 kWh/day
PV share of 1,500 kWh/day hub	12 – 30 %



Route hub — solar canopy, DC posts and BESS behind one connection point.



Bay marking, lighting and safe cable management are part of the design, not an extra.



System architecture — grid, PV, storage, CCS2 dispensers and the charging management system.

Storage does not replace a connection study

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Swap in under five minutes

Riders earn by the trip, so charging time is lost income. Swap moves the charging off the rider and onto the cabinet. Phase 1 runs on one validated 2.56 kWh Energy Module.



Twelve-bay swap cabinet — the standard urban configuration.



Bay module — one door, one pack, blind-mate power and comms.



Sited where riders already stop — petrol stations, stages and shops.

Automated Boda Hub configurations

CONFIGURATION	DOORS	STORED ENERGY	TYPICAL SITE
Compact	8	~20 kWh	Shops, neighbourhood stages
Standard	12	~30 kWh	Petrol stations, urban hubs
Locker bank	24+	Two or more cabinets	Fleet yards, busy corridors



Locker family — compact, standard and bank configurations share one bay module.

Status and limits

Self-service lockers are in design and testing. Live Boda sites on Charging Hub (/hub) may be partner- or staff-operated. Weather rating is under test. Other battery packs only after approval gates. Tariffs and payback come from pilots, not projections.

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One module. Three scales.

The modular energy platform is a design programme, not a product line. Targets below are engineering intent for review — nothing here is certified or for sale.



P1 Go

Portable · conceptual

~1 kWh carry unit. Foldable 100–200 W panel on XT60, UK 3-pin sockets, USB-C and USB-A. Type 2 trickle only — not daily EV charging.



P2 Home

1–4 modules · conceptual

Floor-standing tower beside the consumer board. 2.6 / 5.1 / 7.7 / 10.2 kWh nameplate tiers, 2.4 or 4.8 kW inverter, essential-load sub-board.



Mini Stack

2–6 modules · conceptual

Outdoor SME backup on a plinth with wall shade and service door. Fridge, till and lighting continuity. IP rating not certified yet.

Energy Module — engineering targets

PARAMETER	TARGET
Nominal energy	2.56 kWh
Configuration	16s LiFePO4 · 51.2 V · 50 Ah
Operating voltage	44.8 – 58.4 V
Max discharge	50 A (1.0 C) → 2.56 kW
Max charge	30 A (0.6 C) → ~1.5 kW
Peak discharge	100 A / 10 s [VALIDATE]
Usable AC (backup)	2.17 kWh per module
Daily cycling AC	1.68 kWh per module
Operating temperature	−5 °C to +50 °C, derated above 40 °C
Cycle life target	≥ 4,000 cycles to 80 % SOH at 25 °C, 0.5 C
Enclosure	IP20 indoor · IP55 outdoor rack [CERT]
Communications	CAN 2.0B 500 kbit/s, RS-485 fallback
Mass target	~ 30 kg



The 2.56 kWh Energy Module — the repeated unit of capacity across the platform.

P2 Home stack tiers

MODULES	NAMEPLATE kWh	USABLE AC kWh	BACKUP EXAMPLE
1	2.56	2.17	~4.3 h at 500 W
2	5.12	4.34	~4.3 h at 1 kW
3	7.68	6.51	~3.3 h at 2 kW
4	10.24	8.68	~4.3 h at

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Do not quote nameplate

Usable AC energy is what the customer experiences: nameplate × SOC window × conversion. Quote expected usable kWh and state the reserve. Modular energy is conceptual — not on sale.

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Project-engineered storage, not a catalogue item

Precifarm designs, integrates and develops MegaPack projects. Precifarm does not manufacture cells, power conversion or containers. Every figure below is scoped per site.



MegaPack — container-scale storage for industrial, hub and grid-connected sites.



Storage behind an EV hub — solar, grid and BESS feeding high-power chargers.

Scale stages

STAGE	ENERGY	FORM	TYPICAL USE
Energy Block	Sub-MWh	Single skid or container	Site backup, peak shaving
MegaPack	~5 – 100 MWh	Container array	Industrial, EV hub, C&I
Energy Hub	~25 – 500 MWh	Multi-array site	Grid support, large campus
Energy Park	500 MWh – GWh	Utility plant	Utility firming, market services

Illustrative project shapes

CASE	SIZING	ROLE	EVIDENCE
EV charging hub	1 MW grid · 1 MW solar · 5 MWh BESS	1–2 MW chargers	Illustrative
Industrial site	2 MW / 8 MWh (~4 h)	Peak shaving, continuity	Illustrative
Utility plant	50 MW / 200 MWh (~4 h)	Firming and market services	Illustrative

Licensing

Export and public EV charging may require licences. Storage does not replace grid interconnection studies. Chemistry is LFP where appropriate, confirmed per project.

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The numbers every Precifarm design starts from

Use these for first-pass sizing. Confirm supply, tariff and irradiance on site before quoting.

Kenya design basis

PARAMETER	VALUE
Grid	230 V · 50 Hz · Type G
Design solar (Nairobi)	4.7 PSH · ~3.5 kWh/kWp/day
PV performance ratio	0.75 planning
Canopy PV yield band	4.5 – 5.5 kWh/kWp/day
Grid cost (all-in)	KES 20 – 28 / kWh
E-mobility tariff	~KES 8 off-peak · 16 peak, before levies
Domestic net-metering cap	10 kW
Average monthly outage	~8.4 h reported
Typical Nairobi day	~60 km ≈ ~10 kWh
EV consumption default	~0.17 kWh/km (~10 kWh at 60 km planning)

Precifarm Agent calculation tools

DOMAIN	TOOL
EV energy	calculateEvEnergy()
Charging time	calculateChargingTime()
Peak load	calculatePeakLoad()
Backup sizing	sizeBattery()
Tariff cost	calculateEnergyCost()
Output labels	calculated · estimated · assumed

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Core sizing formulas

QUANTITY	FORMULA	WORKED VALUE
Daily EV energy	E_day = distance × consumption	`60 km × 0.167 ≈ 10 kWh`
Charge time	t = E_day ÷ P_charger	`14.4 ÷ 7 = 2.1 h`
Minimum charger power	P_min = E_day ÷ window	`18 ÷ 10 = 1.8 kW`
Usable storage	E_use = nameplate × SOC window × conversion	`2.56 × 0.90 × 0.94 = 2.17 kWh`
Backup duration	t = E_use ÷ load	`8.68 ÷ 2 = 4.3 h`
PV daily yield	E_pv = kWp × PSH × PR	`50 × 4.7 × 0.75 ≈ 176 kWh/day`
Peak site power	P = E_session ÷ session time	`120 kWh ÷ 0.5 h = 240 kW`
BESS gap	E_gap = (P_demand – P_import) × time	`(253 – 180) × 0.5 ≈ 37 kWh`

Order of work

Establish the daily energy first, then the available charging window, then the minimum power. Only after that pick a product. Sizing the charger before the window is the most common error.

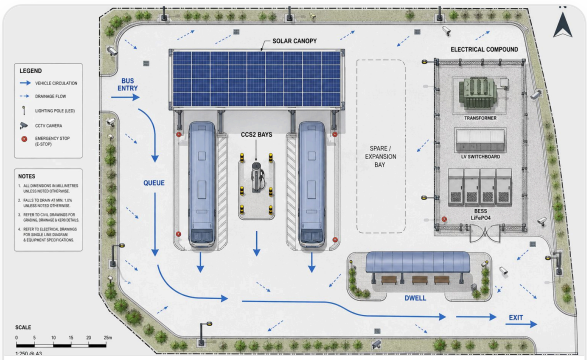
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From load schedule to commissioning

The same sequence applies to a single wallbox and a highway hub. Only the magnitudes change.

Electrical design checklist

- **Load schedule**
  List every circuit with power and duty before sizing anything.
- **Peak demand**
  Apply diversity to household loads; charger load is not diversified.
- **Supply capacity**
  Compare peak demand against main fuse and Kenya Power connection.
- **Cable sizing**
  Size on current-carrying capacity, then verify voltage drop over the run.
- **Protection**
  Overcurrent plus RCD Type A with DC detection, or Type B where required.
- **Distribution board**
  Confirm spare ways, busbar rating and labelling before install day.
- **Earthing**
  Verify arrangement and electrode resistance; assess PME export.
- **Load management**
  Where supply is tight, limit charger current rather than upgrading blindly.



Site plan — bays, clearances, cable routes and equipment positions agreed before civils.



Hub context — lighting, shade and amenity decide whether drivers actually stop.

Kenya approval and hold points

AUTHORITY	SCOPE	WHEN
Kenya Power	Load application, connection study, metering	Before irreversible spend
EPRA	Electrical installation compliance	Design and install
NEMA	Environmental screening where applicable	Site development
County fire	Fire safety approval for public sites	Before public operation
Precifarm CSMS	OCPP registration and M-Pesa test session	Commissioning

Hold points are not paperwork

Each one can change the design. Clear the Kenya Power load application and tariff position before ordering equipment, not after.

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What surrounds the hardware

Charging Hub (/hub) lists open and coming-soon bays. Training keeps sites safe on live hardware. Precifarm Agent speeds first-pass sizing (web workspace + OpenCode). Lipa Pole Pole unlocks Pulse and Pod purchases.

Training tiers

TIER	DURATION	AUDIENCE	VALIDITY
T1	1 day	Hub staff, drivers, site hosts	24 months
T2	3 days	Field technicians	Annual refresher
T3	5 days	Commissioning engineers	Annual refresher + authorisation



Lipa Pole Pole — M-Pesa instalments for Pulse and Pod. Financing, not a charger.



Modular energy family — conceptual renders showing design intent only.

Training delivery

Cohorts of 6 to 16, Nairobi classroom plus field modules on the Nairobi–Kisumu corridor. Certificates issue within five business days. T1 is valid 24 months; T2 and T3 need annual refreshers.

Precifarm Products & Services — Engineering Design Reference
PF-PRODUCT-ENG-001 · v1.1 · 16 October 2026 · Companions: PF-ENG-SOLAR-HUB-001 · PF-MODENERGY-002
Planning basis only. Every installation requires a site survey, Kenya Power confirmation and licensed sign-off. Conceptual products are not offered for sale.

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Precifarm
Electric mobility infrastructure for Africa

Products & Services — Engineering Design Reference

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